

Section A: Pure Mathematics

- 1 Find the set of values of k for which the line $x + y = k$ does not cut the curve

$$y = \frac{5}{x} + x, \text{ leaving your answer in exact form.}$$

[5]

$$x + y = k \Rightarrow y = k - x$$

$$y = \frac{5}{x} + x$$

At the points of intersection,

$$k - x = \frac{5}{x} + x$$

$$\Rightarrow kx - x^2 = 5 + x^2 \Rightarrow 2x^2 - kx + 5 = 0$$

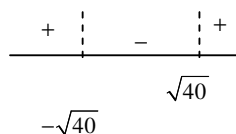
If the line does not cut the curve, there will be no real solutions for the equation $2x^2 - kx + 5 = 0$.

This implies that : $(-k)^2 - 4(2)(5) < 0$

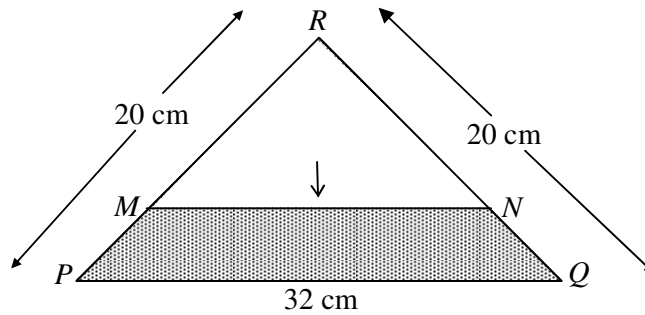
$$\Rightarrow k^2 - 40 < 0$$

$$\Rightarrow (k - \sqrt{40})(k + \sqrt{40}) < 0$$

$$\Rightarrow -\sqrt{40} < k < \sqrt{40}$$

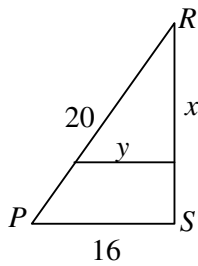


Set of values of k is $\{k \in \mathbb{R} : -\sqrt{40} < k < \sqrt{40}\}$



The diagram shows an isosceles triangle PQR , where $PR = QR = 20$ cm and $PQ = 32$ cm. MN moves towards PQ at a constant speed of 0.6 cm s^{-1} , while keeping parallel to PQ . If MN is x cm from R , show that $MN = \frac{8}{3}x$ cm. [2]

Hence find the rate at which the area of the shaded region is decreasing when MN is halfway towards PQ from R . [4]



By Pythagoras' Theorem,

$$RS^2 = 20^2 - 16^2 = 144$$

$$RS = 12$$

By similar triangles,

$$\frac{y}{x} = \frac{16}{12} = \frac{4}{3}$$

$$y = \frac{4}{3}x$$

$$MN = 2y = \frac{8}{3}x \text{ cm}$$

Let the area of the shaded region be $A \text{ cm}^2$.

$$A = \frac{1}{2}(32)(12) - \frac{1}{2}\left(\frac{8}{3}x\right)x = 192 - \frac{4}{3}x^2$$

$$\frac{dA}{dx} = -\frac{8}{3}x$$

$$\frac{dA}{dt} = \frac{dA}{dx} \cdot \frac{dx}{dt} = \left(-\frac{8}{3}x\right)(0.6) = -1.6x$$

$$\text{When } x = 6, \frac{dA}{dt} = (-1.6)(6) = -9.6.$$

The area of the shaded region is decreasing at a rate of $9.6 \text{ cm}^2 \text{ s}^{-1}$.

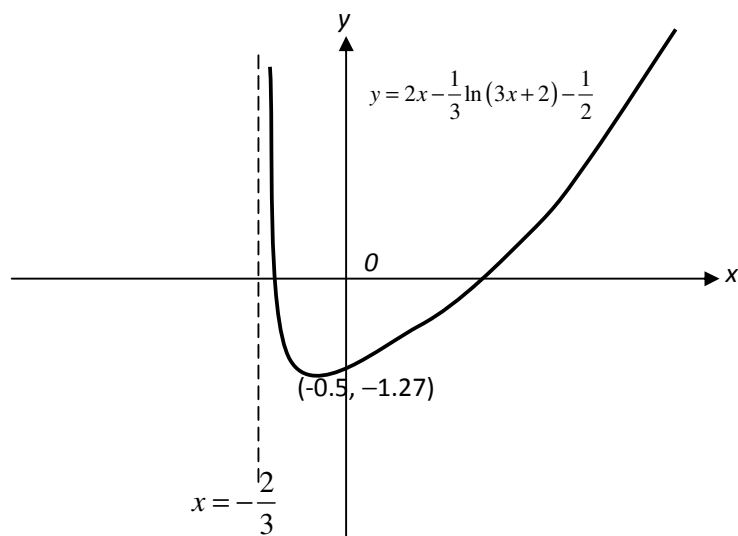
- 3 The gradient function of a curve C at any point is given by the expression

$$\frac{dy}{dx} = 2 - \frac{1}{3x+2}.$$

- (i) Find the equation of the normal to the curve at the point $P\left(-\frac{1}{3}, -\frac{7}{6}\right)$. Express your answer in the form $ax+by+c=0$, where a , b and c are positive integers. [3]
- (ii) Find the equation of the curve C . [3]
- (iii) Sketch the curve C , stating the equations of any asymptotes and the coordinates of any stationary points. [3]

(i)	$\frac{dy}{dx} = 2 - \frac{1}{3x+2}$ At $P\left(-\frac{1}{3}, -\frac{7}{6}\right)$, $\text{Gradient of normal} = \frac{-1}{2 - \frac{1}{3\left(-\frac{1}{3}\right) + 2}} = -1$ $y - \left(-\frac{7}{6}\right) = -\left(x - \left(-\frac{1}{3}\right)\right)$ $y = -x - \frac{1}{3} - \frac{7}{6}$ $y = -x - \frac{3}{2}$ $2y + 2x + 3 = 0$
(ii)	$\frac{dy}{dx} = 2 - \frac{1}{3x+2}$ $y = 2x - \frac{1}{3} \ln(3x+2) + c$ At $P\left(-\frac{1}{3}, -\frac{7}{6}\right)$, $-\frac{7}{6} = 2\left(-\frac{1}{3}\right) - \frac{1}{3} \ln\left(3\left(-\frac{1}{3}\right) + 2\right) + c$ $c = -\frac{7}{6} + \frac{2}{3} = -\frac{1}{2}$ $y = 2x - \frac{1}{3} \ln(3x+2) - \frac{1}{2}$

(iii)



- 4 An open rectangular gift box of height h cm has a rectangular base of sides x cm and $3x$ cm. If the volume of the gift box is 600 cm^3 ,

(i) show that the total surface area, A , is $3x^2 + \frac{1600}{x}$, [3]

(ii) use differentiation to find the value of x for which A is a minimum. [4]

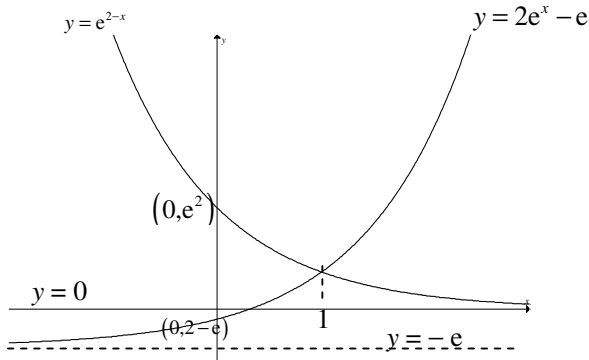
(i)	$V = 3x^2 h$ $h = \frac{200}{x^2}$ $A = 3x^2 + 2(3xh) + 2(xh)$ $A = 3x^2 + 6x \frac{200}{x^2} + 2x \frac{200}{x^2}$ $A = 3x^2 + \frac{1600}{x}$ (shown)								
(ii)	$\frac{dA}{dx} = 6x - \frac{1600}{x^2}$ For stationary point, $\frac{dA}{dx} = 6x - \frac{1600}{x^2} = 0$ $6x^3 = 1600$ $x = \sqrt[3]{\frac{800}{3}}$ $x = 6.44$ <table><tr><td>x</td><td>$\left(\sqrt[3]{\frac{800}{3}}\right)^{-}$</td><td>$\sqrt[3]{\frac{800}{3}}$</td><td>$\left(\sqrt[3]{\frac{800}{3}}\right)^{+}$</td></tr><tr><td>Slope of $\frac{dA}{dx}$</td><td></td><td></td><td></td></tr></table> Therefore $x = 6.44$ gives minimum A.	x	$\left(\sqrt[3]{\frac{800}{3}}\right)^{-}$	$\sqrt[3]{\frac{800}{3}}$	$\left(\sqrt[3]{\frac{800}{3}}\right)^{+}$	Slope of $\frac{dA}{dx}$			
x	$\left(\sqrt[3]{\frac{800}{3}}\right)^{-}$	$\sqrt[3]{\frac{800}{3}}$	$\left(\sqrt[3]{\frac{800}{3}}\right)^{+}$						
Slope of $\frac{dA}{dx}$									

- 5 (i) By means of the substitution $u = e^x$, show algebraically that $x = 1$ is a root of the equation

$$2e^x = e + e^{2-x}. \quad [3]$$

- (ii) On a single diagram, sketch the graphs of $y = 2e^x - e$ and $y = e^{2-x}$, stating clearly the exact coordinates of any points of intersection with the y -axis and the equations of the asymptotes. [2]

- (iii) Find the exact area of the region bounded by the curves $y = 2e^x - e$, $y = e^{2-x}$ and the line $x = 2$, giving your answer in terms of e . [3]

(i)	$2e^x = e + e^{2-x}$ $2e^{2x} = e \cdot e^x + e^2$ $2e^{2x} - e \cdot e^x - e^2 = 0$ Let $u = e^x$. $2u^2 - eu - e^2 = 0$ $(u - e)(2u + e) = 0$ $u = e \quad \text{or} \quad u = -\frac{1}{2}e$ $e^x = e \quad \text{or} \quad e^x = -\frac{1}{2}e \quad (\text{no soln as } e^x > 0)$ $x = 1$
(ii)	
(iii)	$\text{Area of region} = \int_1^2 [(2e^x - e) - e^{2-x}] dx$ $= [2e^x - ex + e^{2-x}]_1^2$ $= (2e^2 - 2e + 1) - (2e^1 - e + e)$ $= 1 + 2e^2 - 4e$

Section B: Statistics [60 marks]

- 6** The events H and K are such that that $P(H)=h, P(K)=k, P(H \cap K)=0.03$ and $P(H \cup K)=0.51$.

(i) Find $P(H' \cap K')$. [1]

(ii) Given that the events H and K are independent, find a quadratic equation satisfied by k and hence find the possible values of $P(K)$. [4]

(i)	$P(H' \cap K') = 1 - P(H \cup K) = 0.49$
(ii)	$P(H \cup K) = P(H) + P(K) - P(H \cap K)$ $= h + k - hk$ $P(H \cap K) = hk$ $0.03 = hk$ $h = \frac{0.03}{k}$ $0.51 = \frac{0.03}{k} + k - \frac{0.03}{k}k$ $0.54 = \frac{0.03}{k} + k$ $k^2 - 0.54k + 0.03 = 0$ $k = 0.477 \text{ or } k = 0.0629$ $P(K) = 0.477 \text{ or } 0.0629$

- 7 In a certain large town, it is found that 76% of people wear a watch on their left wrist, 15% wear a watch on their right wrist and 9% do not wear a watch.
- (i) A random sample of 14 adults is taken. Find the probability that more than 2 but at most 8 adults do not wear a watch. [2]
- (ii) A random sample of 200 adults is taken. Use a suitable approximation to estimate the probability that at least 155 wear a watch on their left wrist. You should state the mean and variance of the distribution used in the approximation. [4]
- (iii) A random sample of 12 adults is taken. Find the most likely number of adults who wear a watch on their right wrist in the sample. [1]

7 (i)	Let X be the number of adults, out of 14, who did not wear a watch. Then $X \sim B(14, 0.09)$. $P(2 < X \leq 8) = P(X \leq 8) - P(X \leq 2)$ $= 0.12551 = 0.126 \text{ (3 s.f.)}$
(ii)	Let Y be the number of adults, out 200, who wore a watch on their left wrist. Then $Y \sim B(200, 0.76)$. Since $n = 200$ (sufficiently large), $p = 0.76$ such that $np = 152 (> 5)$, $nq = 48 (> 5)$ } $Y \sim N(np, npq)$ approximately. i.e., $Y \sim N(152, 36.48)$ approximately. $P(Y \geq 155) = P(Y > 154.5) \text{ (with continuity correction)}$ $= 0.339 \text{ (3 s.f.)}$
(iii)	Let W be the number of adults, out of 12, who wore a watch on their right wrist. $W \sim B(12, 0.15)$ From GC, $P(W = 0) = 0.14224$ $P(W = 1) = 0.30122$ $P(W = 2) = 0.29236$ Hence required mostly likely number is 1.

- 8 (a) The editor of a magazine wishes to obtain the views of a random sample of readers about the future of the magazine.
- (i) A sub-editor proposes that they include in one issue of the magazine a questionnaire for readers to complete and return. Give a reason why the readers who return the questionnaire would not form a random sample. [1]
 - (ii) The editor decides to select a random sample of 120 readers from the 7320 regular readers. Describe how a systematic sample could be carried out. [2]
- (b) The editor claims that the average amount of money spent by a reader on magazines per month is \$14. A random sample of 60 readers is taken and were asked the amount (\$x), correct to the nearest dollars, they spent on magazines during the previous month. The results are summarised by
- $$\sum (x-14) = -78 \text{ and } \sum (x-14)^2 = 1232.$$
- (i) Find the unbiased estimate of the population variance. [1]
 - (ii) Test, at the 4% level, whether the editor is overstating the average amount of money spent by a reader on magazines per month. [4]

8 (a)	Possible reasons:
(i)	Only the more committed or less busy will complete and return; Only readers of that particular issue have the chance to participate in the survey.
(ii)	Sampling interval, $k = \frac{7320}{120} = 61$ Get a namelist of the 7320 regular readers and arranged in alphabetical order. Choose the first reader <u>at random</u> from the first 61 names. Choose every 61 st name after the first name has been chosen until a sample of 120 is obtained.
(b)	unbiased estimate of population variance,
(i)	$s^2 = \frac{1}{59} \left[1232 - \frac{78^2}{60} \right] \approx 19.1627 = 19.2 \text{ (3 s.f.)}$
(ii)	Let μ be the average amount of money spent by a reader on magazines per month. $H_0 : \mu = 14$ $H_1 : \mu < 14$ Level of significance: 4% Under H_0 , test statistics, $Z = \frac{\bar{X} - 14}{s / \sqrt{60}} \sim N(0,1)$ approximately by C.L.T. $\bar{x} = \frac{\sum (x-14)}{60} + 14 = \frac{-78}{60} + 14 = 12.7$

	From GC, $z = -2.3003$ $p\text{-value} = 0.010715 (< 0.04)$ Reject H_0. There is significant evidence at the 4% level of significance that the editor is overstating his claim.
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9 The lifespan, in years, of a certain type of halogen bulbs follows a normal distribution with mean 2 and standard deviation 0.6.

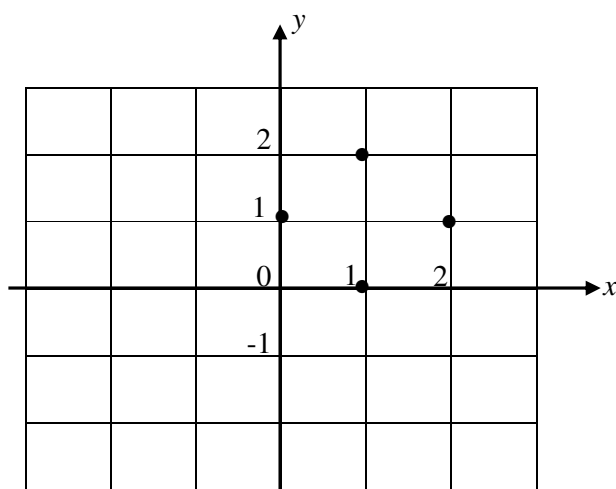
- (i) Find the probability that the lifespan of a randomly chosen halogen bulb is between 1.9 and 2.5 years. [1]
- (ii) Twenty bulbs was randomly taken. Find the probability that at most 3 of them have a lifespan of between 1.9 and 2.5 years. [2]
- (iii) A random sample of n halogen bulbs is taken. Find the smallest value of n such that there is a probability of at most 0.05 that the mean lifespan of these halogen bulbs exceeds 2.2 years. [5]

(i)	Let X hours be the lifespan of a halogen bulb $X \sim N(2, 0.6^2)$ $P(1.9 < X < 2.5) = 0.36386 = 0.364$
(ii)	Let Y be rv no of halogen bulbs that have lifespan less than 1.8 hours out of 20. $Y \sim B(20, 0.36386)$ $P(Y \leq 3) = 0.03391 = 0.0339$
(iii)	$\bar{X} = \frac{X_1 + \dots + X_n}{n} \sim N\left(2, \frac{0.6^2}{n}\right)$ $P(\bar{X} > 2.2) \leq 0.05$ $P(\bar{X} < 2.2) \geq 0.95$ $P\left(Z < \frac{2.2 - 2}{\frac{0.6}{\sqrt{n}}}\right) \geq 0.95$ $\frac{2.2 - 2}{\frac{0.6}{\sqrt{n}}} \geq 1.6448$ $n \geq 24.35$ Least $n = 25$

- 10 (a)** John sells ice cream at a park. The mid-day temperature, x , in degree Celsius ($^{\circ}\text{C}$) over a period of 8 days, together with the daily profits, y , from the sale of the ice cream are given in the following table.

x	32.1	10.2	23.5	12.7	38.6	20.3	27.8	17.4
y	128	47	98	56	152	83	111	78

- (i) Give a sketch of the scatter diagram for the data as shown on your calculator. [2]
 - (ii) Obtain the equation of the regression line of y on x in the form $y = a + bx$. Interpret the values of a and b in the context of this question. [3]
 - (iii) Calculate an estimate of the profit when the mid-day temperature is 35°C . Comment on the reliability of this estimate. [2]
 - (iv) Explain why the use of the regression line of x on y would not be appropriate in part (iii). [1]
- (b)** The scatter diagram shows a sample of four pairs of values (x, y) . Give the coordinates of a fifth point such that the linear (product moment) correlation coefficient for all five points is
- (i) zero, [1]
 - (ii) negative. [1]



(i)	
(ii)	LinReg $y=a+bx$ $a=10.49024012$ $b=3.664173489$ $r^2=.9973789303$ $r=.9986886053$ $y = 10.490 + 3.6642x$ $y = a + bx$ When the midday temp is 0°C, the profit is \$10.5 For every increase of 1°C in the midday temp, the profit will increase by \$3.66
(iii)	$y = 10.490 + 3.6642(35)$ $y = 138.737 = 139$ Therefore, the profit is \$139. Since $x = 35$ is within the range of x and $r = 0.998$, therefore the estimate is reliable
(iv)	regression line of x on y is not appropriate because x is an independent variable .
b(i)	(1,1)
(ii)	(0,2)

- 11** A coin is known to be biased in such a way that, when it is tossed, the probability of a head showing is $\frac{1}{3}$. In a probability experiment, two boxes have the following contents:

A red box contains 2 white counters and 4 black counters.

A yellow box contains 3 white counters and 5 black counters.

The biased coin is tossed once. If the biased coin shows a head, two counters are taken from the red box, at random and without replacement. If the biased coin shows a tail, two counters are taken from the yellow box, at random and without replacement. Events H and W are defined as follows.

H : The biased coin shows a head

W : One white counter and one black counter are obtained.

Show that $P(W \cap H) = \frac{8}{45}$. [2]

Find, giving each of your answers as an exact fraction in its lowest terms,

(i) $P(H | W)$, [3]

(ii) $P(W \cup H)$. [2]

All the counters are now placed in a single box. Four counters are taken at random, each counter being replaced before the next one is taken. Find the probability that there is at least one counter of each colour, giving your answer correct to 3 significant figures. [3]

	$P(W \cap H) = \frac{1}{3} \times \frac{2}{6} \times \frac{4}{5} \times 2 = \frac{8}{45}$
(i)	$P(H W) = \frac{P(H \cap W)}{P(W)}$ $= \frac{\frac{8}{45}}{\frac{8}{45} + \frac{2}{3} \times \frac{3}{8} \times \frac{5}{7} \times 2}$ $= \frac{112}{337}$
(ii)	$P(W \cup H)$ $= P(W) + P(H) - P(W \cap H)$ $= \frac{8}{45} + \frac{2}{3} \times \frac{3}{8} \times \frac{5}{7} \times 2 + \frac{1}{3} - \frac{8}{45}$ $= \frac{29}{42}$
	Probability that there is at least one counter of each colour $= 1 - P(\text{all white}) - P(\text{all black})$ $= 1 - \left[\left(\frac{5}{14} \right)^4 + \left(\frac{9}{14} \right)^4 \right]$ $= 0.813$

- 12** A machine makes metal rods. The diameters of the rods are normally distributed with mean 1.00 cm and standard deviation σ cm. A rod is oversized if its diameter exceeds 1.05 cm. It is found that from experience that 1% of the rods produced by the machine are oversized. Find the value of σ , giving your answer correct to 3 significant figures.

[3]

Another machine makes metal rings. The internal diameters of the rings are normally distributed with mean 1.02 cm and standard deviation 0.04 cm.

- (i) Find the probability that the sum of the internal diameters of 5 randomly chosen rings exceeds 5.3 cm. [2]
- (ii) Find the probability that the sum of the internal diameters of 5 randomly chosen rings is less than five times the internal diameter of a randomly chosen rod. [3]
- (iii) A random sample of 30 metal rings is taken and the sample mean internal diameter is found to be $\bar{x}$ cm. A test is carried out, at the 10% significance level, to determine whether the mean internal diameter of a ring is 1.02 cm. Find the set of values of $\bar{x}$ for which the result of the test would be to reject the null hypothesis. [4]

12	Let X cm be the internal diameter of a rod. $X \sim N(1, \sigma^2)$ $P(X > 1.05) = 0.01$ $\Rightarrow P(X < 1.05) = 0.99$ $\Rightarrow P\left(Z < \frac{1.05 - 1}{\sigma}\right) = 0.99$ $\Rightarrow \frac{0.05}{\sigma} = 2.326$ $\Rightarrow \sigma = 0.021496 = 0.0215$ (3 s.f.) (i) Let Y cm be the internal diameter of a ring. $Y \sim N(1.02, 0.04^2)$ $Y_1 + Y_2 + \dots + Y_5 \sim N(5 \times 1.02, 5 \times 0.04^2)$ $= N(5.1, 0.008)$ Required probability = $P(Y_1 + Y_2 + \dots + Y_5 > 5.3)$ $= 0.0126736 = 0.0127$ (3 s.f.) (ii) $Y_1 + Y_2 + \dots + Y_5 - 5X \sim N(5.1 - 5(1), 0.008 + 5^2 \times 0.0215^2)$ $= N(0.1, 0.019556)$ Required probability = $P(Y_1 + Y_2 + \dots + Y_5 - 5X < 0)$ $= 0.237$ (3 s.f.)
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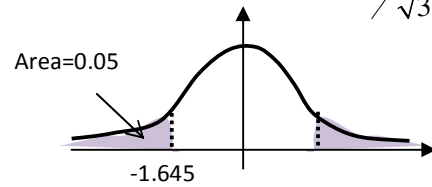
(iii) Let μ cm be the mean internal diameter of a ring.

$$H_0 : \mu = 1.02$$

$$H_1 : \mu \neq 1.02$$

Level of significance: 10%

Under H_0 , test statistics, $Z = \frac{\bar{X} - 1.02}{0.04/\sqrt{30}} \sim N(0,1)$



Reject H_0 if $z \leq -1.645$ or $z \geq 1.645$

$$\frac{\bar{x} - 1.02}{\frac{0.04}{\sqrt{30}}} \leq -1.645 \text{ or } \frac{\bar{x} - 1.02}{\frac{0.04}{\sqrt{30}}} \geq 1.645$$

$$\bar{x} \leq 1.00799 \text{ or } \bar{x} \geq 1.03201$$

$$\bar{x} \leq 1.01 \text{ or } \bar{x} \geq 1.03$$